

Housing Market Intelligence Power BI Dashboard

MS SQL Server -> Power BI -> DAX -> Incremental Refresh -> Row-Level Security

100K

housing transactions

192.6bn

total sales value

32+ yrs

1992-2024 history

Secure BI

RLS + incremental refresh

Housing Market Intelligence Dashboard

Industry-Standard Project Report | MS SQL Server -> Power BI

Executive portfolio summary

Converted 100,000 housing transactions into a governed Power BI analytics product with SQL Server sourcing, page-level DAX measure tables, incremental refresh, and role-level security. The project answers business questions around regional revenue, premium markets, property pricing, offer-to-purchase alignment, and sales-type performance.

100,000

Transactions

192.6bn

Total Sales Value

1992-2024

Date Coverage

13.7bn

Last 12M Sales

1. Executive Summary

This project turns historical housing transactions into a business-ready intelligence product. It combines exploratory analysis, SQL Server sourcing, Power BI semantic modeling, DAX measures, production refresh design, and role-level security into one cohesive analytics solution.

The dashboard is designed for executive and regional decision makers who need to understand market size, revenue concentration, premium regions, city-level opportunities, property-type economics, and offer-to-purchase behavior.

Theme	Report Summary
Business objective	Create a secure Power BI dashboard that explains housing sales performance and pricing behavior.
Data foundation	100,000 transactions across 19 original fields, covering 1992-01-05 to 2024-10-24.
BI stack	MS SQL Server, Power BI Desktop, DAX, Power BI Service, incremental refresh, RLS.
Primary users	Executives, regional managers, pricing teams, and analytics stakeholders.
Core outcome	Actionable insights on regional revenue, premium pricing, sales-type movement, property mix, and city focus.

Data storytelling anchor

The business story is simple: revenue is concentrated in a few geographies, premium pricing is not the same as volume, offer prices are highly aligned with final purchase prices, and secure self-service reporting lets each audience view the slice of the market they are allowed to manage.

2. Business Questions Answered

#	Business question	Answer / insight	Business impact
1	How large is the market?	100K transactions and 192.6bn total sales value.	Creates a reliable baseline for leadership reviews.
2	Which regions drive revenue?	Zealand: 95.0bn; Jutland: 81.5bn; together over 91% of sales.	Focuses regional strategy and commercial planning.
3	Which markets are premium?	Zealand leads average price per sqm at 20.9K; Bornholm is lowest at 10.6K.	Supports differentiated pricing and positioning.
4	Are offer and purchase prices aligned?	Correlation is about 0.998; 64% close at calculated offer price.	Validates pricing quality and highlights outlier-monitoring opportunities.
5	Which sales types are changing?	Auction is strongest latest YoY, while 2024 sales types are impacted by partial-year data.	Prevents misleading current-year performance interpretation.
6	Which house types are valuable?	Farms have highest average purchase price; apartments have highest price per sqm.	Separates total value from density premium.
7	Which cities deserve focus?	København S, Aarhus C, København A, Roskilde, Silkeborg, Aalborg lead sales value.	Improves local market targeting.
8	How does age affect pricing?	0-15 year homes show the highest median price; 16-45 age segment reduces average price in the key-influencer view.	Informs valuation and renovation strategy.
9	How can this scale securely?	Incremental refresh and RLS support governed Power BI Service deployment.	Improves performance, refresh efficiency, and access control.

3. Data Foundation and Methodology

The dataset contains housing transactions with date, quarter, property type, sales type, build year, purchase price, offer-to-purchase percentage change, rooms, area in square meters, city, area, region, and macroeconomic indicators such as interest, inflation, and mortgage bond yield.

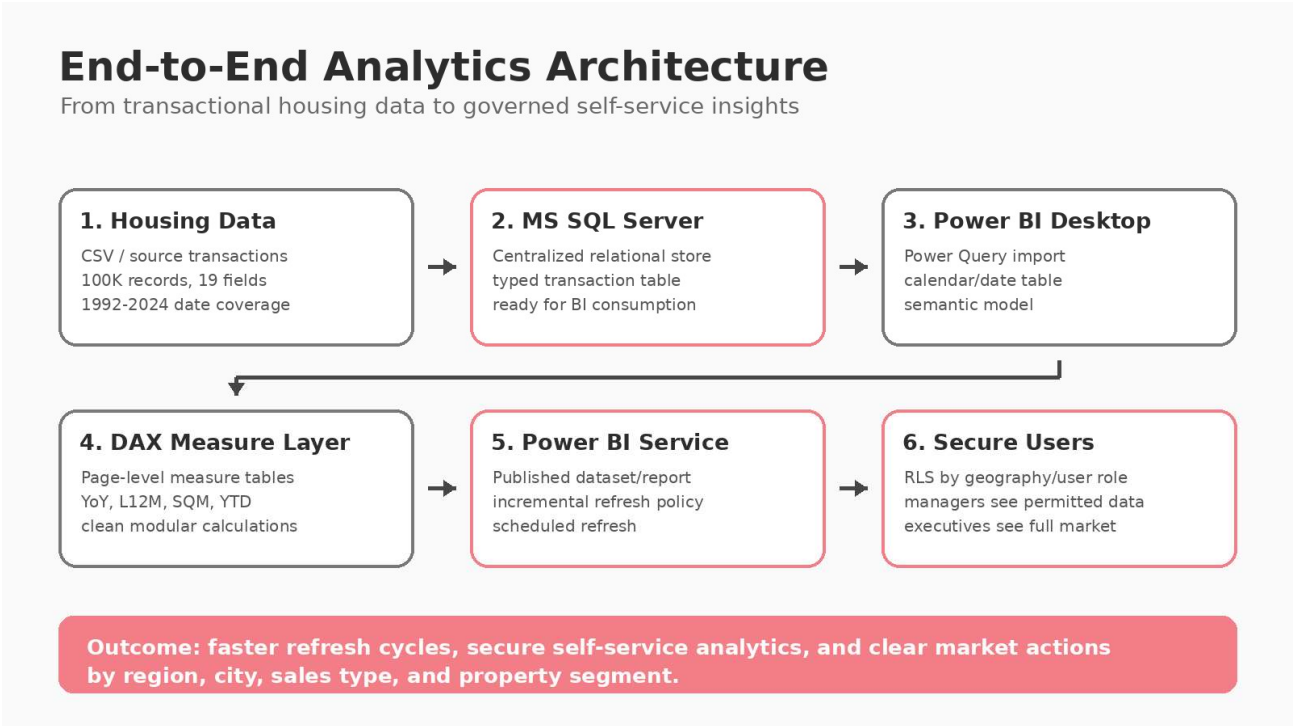
Dataset profile

Rows analyzed	100,000	Original columns	19
Date range	1992-01-05 to 2024-10-24	Duplicate rows	0
Missing macro indicator rows	77	Regions	Zealand, Jutland, Fyn & islands, Bornholm
House types	Villa, Apartment, Summerhouse, Townhouse, Farm	Sales types	regular_sale, family_sale, other_sale, auction

Methodology

Step	What was done	Why it mattered
1. Data ingestion	Housing data stored in MS SQL Server and imported into Power BI.	Creates a realistic BI source layer instead of a file-only dashboard.
2. EDA validation	Python notebook validated shape, missing values, KPIs, region metrics, house types, cities, and age groups.	Ensures dashboard numbers tell a consistent story.
3. Semantic modeling	Date fields, quarter/year/month logic, calculated offer price, and price-per-sqm measures were organized.	Provides a clean calculation layer for self-service reporting.
4. DAX measure design	Measures were grouped into page-level measure tables for overview, sales performance, and property-type analysis.	Improves maintainability and makes the report easier for recruiters and BI teams to review.
5. Power BI Service deployment	Incremental refresh and RLS were configured after publishing.	Turns the dashboard into a governed, scalable BI asset.

4. Solution Architecture



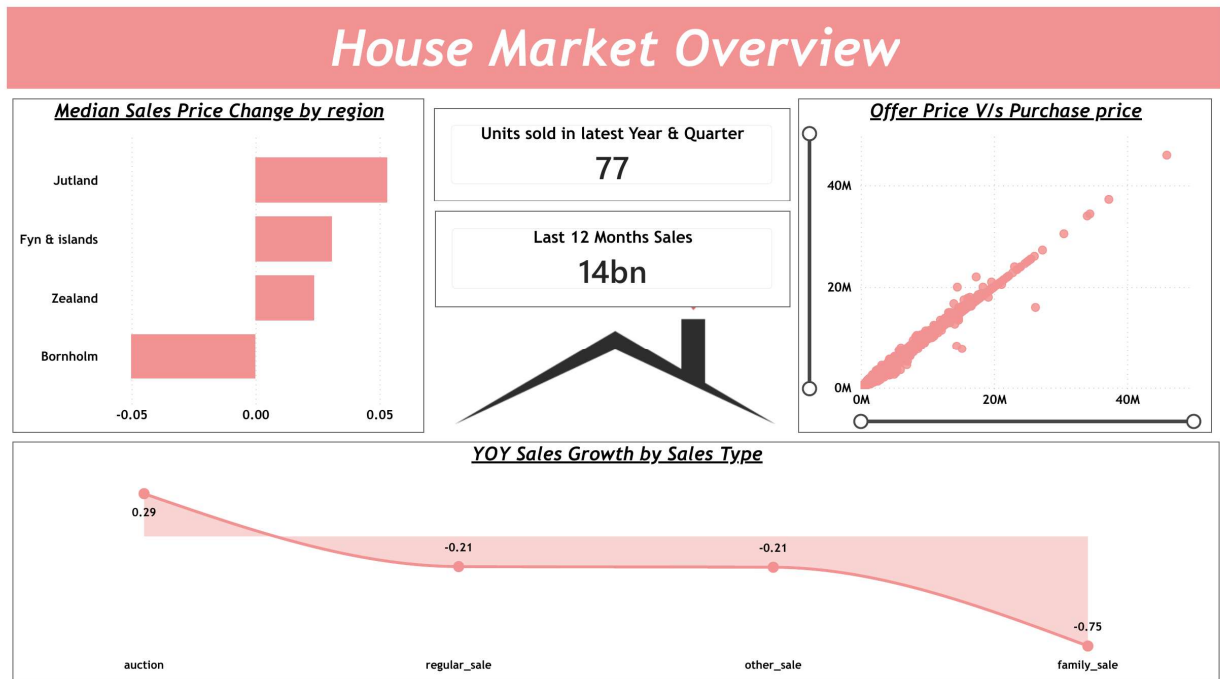
End-to-end solution architecture: data source, SQL Server, Power BI Desktop, DAX layer, Power BI Service, and governed users.

Component	Implementation	Business value
MS SQL Server	Housing data was stored in SQL Server before being imported into Power BI.	Models a real enterprise BI pattern and separates source data from reporting.
Power Query	Imported and prepared data for the model, including date-based logic required for incremental refresh.	Reduces manual data preparation and enables repeatable refresh.
Semantic model	Created a reusable model around transactions, time, geography, property type, and sales type.	Provides one trusted layer for report pages and slicers.
Page-level measure tables	Organized DAX measures separately by dashboard page.	Keeps complex calculations discoverable and maintainable.
Incremental refresh	Configured in Power BI Service using date-based refresh logic.	Avoids full reprocessing of more than 30 years of history on every refresh.
Role-level security	Configured RLS roles to restrict users by permitted data slices such as geography or role.	Enables secure self-service access for managers and executives.

5. Dashboard Storyboard

Page 1 - House Market Overview

This page answers the leadership question: "What is happening in the market right now?" It combines KPI cards with regional price change, offer-vs-purchase alignment, and sales-type YoY movement.

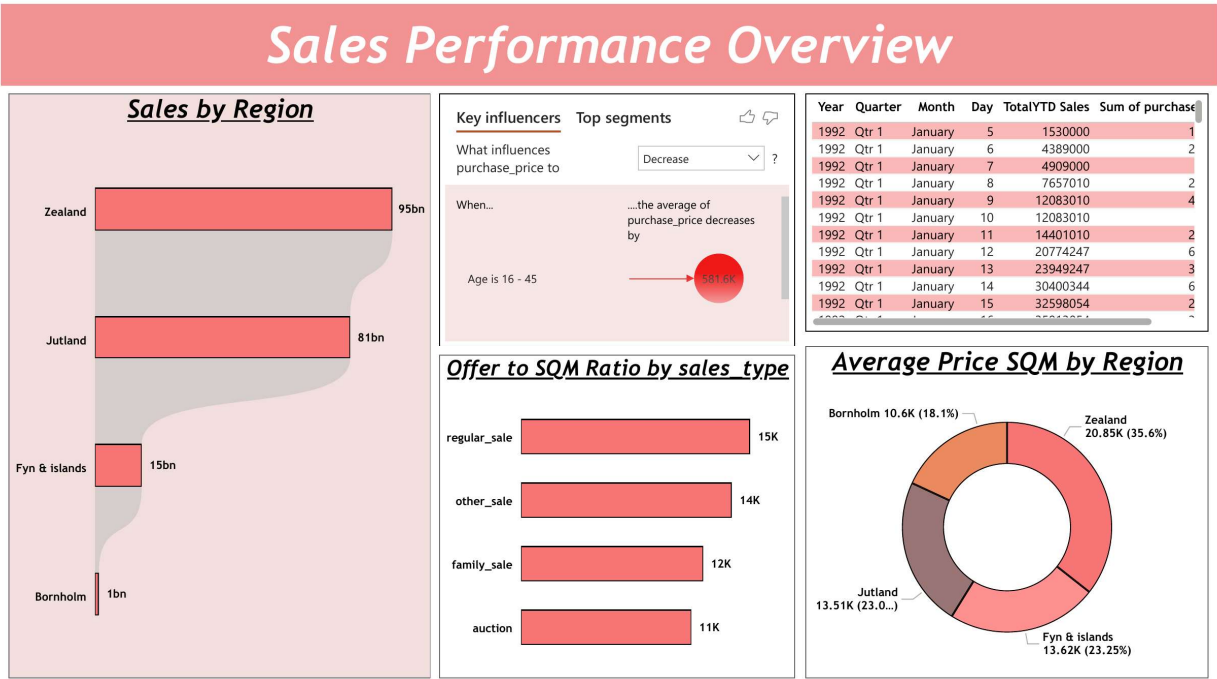


Dashboard page 1: House Market Overview.

- Latest quarter units sold: 77, with the current quarter partial through 2024-10-24.
- Last 12 months sales value: 13.7bn.
- Regional median price change shows Jutland, Fyn & islands, and Zealand increasing, while Bornholm declines.
- Offer-vs-purchase scatter shows a strong diagonal relationship, indicating close alignment between offer and closing prices.

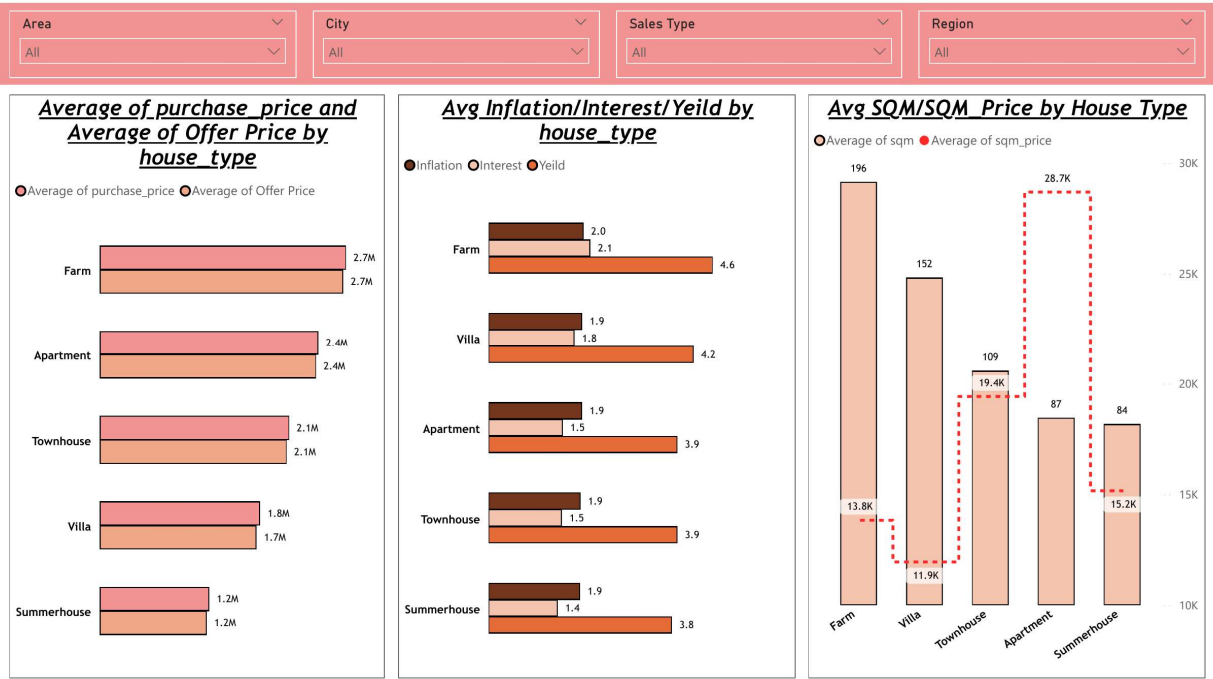
Page 2 - Sales Performance Overview

This page answers: "Where is revenue coming from, and what segments influence price?" It compares regional sales value, price per square meter, offer-to-sqm ratio by sales type, and key influencers of purchase price.



Page 3 - Property Type and Pricing Deep Dive

This page answers: "Which property types drive absolute price and unit economics?" It uses slicers for area, city, sales type, and region while comparing purchase price, offer price, macro indicators, sqm, and sqm price by house type.



Dashboard page 3: Property Type and Pricing Deep Dive.

- Farms have the highest average purchase price and average offer price, near 2.7M.
- Apartments have the highest average sqm price, around 28.7K, showing premium density/location economics.
- Villas provide the largest transaction count and total sales value, making them essential for volume analysis.
- Slicers allow business users to filter by area, city, sales type, and region for localized decision-making.

6. Key Insights and Business Interpretation

Regional market concentration

Region	Sales Value	Sales Share	Transactions	Avg Price/SQM	Interpretation
Zealand	95.0bn	49.3%	39,740	20.9K	Largest revenue and premium price-per-sqm market.
Jutland	81.5bn	42.3%	49,937	13.5K	Highest transaction volume with strong overall revenue.
Fyn & islands	14.9bn	7.7%	9,264	13.6K	Smaller but price-per-sqm similar to Jutland.
Bornholm	1.2bn	0.6%	1,059	10.6K	Smallest and most affordable region.

Business interpretation

Revenue scale and price premium are not the same. Zealand is both the largest revenue region and the premium price-per-sqm market. Jutland is a high-volume, high-revenue region but has a lower average sqm price, which means strategy should differ by geography.

Sales type and pricing behavior

Metric	Result	Interpretation
Latest YoY - auction	+29.1%	Strongest latest YoY growth in the dataset.
Latest YoY - regular_sale	-20.6%	Decline is impacted by partial 2024 data.
Latest YoY - other_sale	-21.1%	Requires monitoring by quarter and region.
Latest YoY - family_sale	-74.8%	Weakest movement; likely segment-specific and affected by partial-year data.
Offer vs purchase correlation	0.998	Offer pricing is highly aligned with final purchase price.
Transactions closing at offer	64.0%	Pricing approach appears stable for most transactions.

Property type economics

House Type	Avg Purchase	Avg Offer	Avg SQM	Avg SQM Price	Strategic read
Farm	2.74M	2.70M	196	13.8K	Largest average ticket size.
Apartment	2.43M	2.40M	87	28.7K	Highest unit economics by sqm.
Townhouse	2.11M	2.08M	109	19.4K	Strong urban middle segment.
Villa	1.79M	1.74M	152	11.9K	Largest transaction and total-sales base.
Summerhouse	1.22M	1.19M	84	15.2K	Lower ticket, lifestyle segment.

7. Business Impact

Impact area	How the project helps the business
Revenue prioritization	Highlights the geographies and cities that drive the largest share of sales value, so leadership can allocate resources more precisely.
Pricing strategy	Compares price per sqm, house type value, and offer-to-purchase behavior to support sharper pricing decisions.
Market segmentation	Separates premium markets, high-volume markets, property types, and age bands into distinct decision segments.
Operational efficiency	Incremental refresh prevents unnecessary refresh of more than 30 years of historical data.
Governance and security	RLS enables self-service reporting while protecting geography- or role-specific access.
Recruiter signal	Demonstrates end-to-end BI capability: SQL, Power BI, DAX, data modeling, security, refresh optimization, and business storytelling.

Recommended business actions

- Treat Zealand as a premium-pricing and executive-watch region because it leads revenue and average price per sqm.
- Monitor Jutland as a high-volume market with meaningful recent median-price movement.
- Build city-level campaigns around København S, Aarhus C, København A, Roskilde, Silkeborg, and Aalborg.
- Use offer-vs-purchase variance as a pricing exception report to identify unusually high or low closing gaps.
- Interpret 2024 sales-type declines carefully because the current-year data is partial through October 24, 2024.
- Extend the model with predictive valuation or lead-scoring once more external economic and listing data is available.

8. Technical Appendix

Representative DAX measure patterns

The report uses modular DAX measures grouped into page-level measure tables. Representative patterns are shown below; measure names can be adapted to the final PBIX naming convention.

Total Sales = SUM ('Housing Data'[purchase_price])
Units Sold = COUNTROWS ('Housing Data')
Last 12 Months Sales = CALCULATE ([Total Sales], DATESINPERIOD ('Date'[Date], MAX ('Date'[Date]), -12, MONTH))
YoY Sales Growth % = DIVIDE ([Total Sales] - [Total Sales PY], [Total Sales PY])
Offer Price = SUMX ('Housing Data', 'Housing Data'[purchase_price] * (1 + 'Housing Data'[%_change_between_offer_and_purchase] / 100))
Offer to SQM Ratio = DIVIDE ([Offer Price], SUM ('Housing Data'[sqm]))

Incremental refresh setup

- Created Power Query parameters such as RangeStart and RangeEnd.
- Filtered the transaction date column using those parameters.
- Published to Power BI Service so the refresh policy could create and maintain partitions.
- Designed the approach to avoid reloading the entire historical dataset at each refresh.

Role-level security setup

- Configured RLS roles in the Power BI model and tested restricted views after publishing.
- Used geography/role filters so regional managers can access only permitted slices while executives can access the full dashboard.
- The model can be extended to dynamic RLS with a user-access table and USERPRINCIPALNAME().

Final takeaway

This project demonstrates a full BI workflow: data sourcing in SQL Server, validation through Python EDA, Power BI semantic modeling, DAX measure engineering, dashboard storytelling, incremental refresh, and role-level security. It is structured to communicate both technical depth and business impact to recruiters and hiring managers.